

VITARA 2.0 -- Executive Summary & Feature Brief

STATUS: LIVE v2.0

Vidyalankar Institute of Technology Academic & Career Acceleration Platform

React 18 | Node.js | Gemini RAG

A unified, next-generation AI-powered institutional ecosystem built to modernize academic ERP compliance, faculty mentorship, institutional RAG intelligence, and real-time skill-gap career acceleration for VIT Mumbai.

14	12	17	75%	<150ms	-60%
Student Tabs	Mentor Modules	Admin Ops Tabs	ERP Cutoff Guard	RAG Query Speed	Bundle Size Cut

1. System Purpose & Core Architectural Pillars

1. Academic ERP & Compliance Live attendance monitor with statutory 75% cutoff warnings, safe-to-miss lecture calculators, semester transcripts, and lab coursework submission queues.	2. AI Mentorship Matcher 4-factor matching algorithm (40% goals, 25% domain, 10% course, 10% dept), real-time approval pipelines, and mentor-assigned university course tracks.	3. Grounded RAG & Roadmaps Grounded Gemini 2.5 Flash / 3.6 Flash RAG over autonomous ordinances & syllabi. Dynamic skill-gap radar (AI, Cloud, Fullstack) with milestone tracking.
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2. Comprehensive Role-Based Feature Modules

Portal / Module	Target Role	Key Features & Functional Capabilities	Impact / Value Delivered
Student Portal 14 Live Tabs	Students (UG / PG)	• Live Attendance & ERP: 75% cutoff threshold, safe-to-miss calculator, makeup requests. • Academic Transcript: CS501-CS503 credits, internal assessments, CGPA trends. • Coursework & Submissions: Drag & drop lab upload with timestamp verification. • AI Mentorship Hub: Request mentors, 1-on-1 booking, mentor feedback notes. • Career Roadmap & Skill Radar: AI/ML, Fullstack, Cloud milestone checklist. • AI Recommended Courses: Mentor-assigned Stanford, MIT, Coursera tracks. • Per-Account Onboarding: First-login career goal & specialization selector.	Automated 75% statutory compliance protection; 3x faster mentor connectivity; industry-aligned skills.
Mentor Portal 12 Live Tabs	Faculty Mentors	• AI Matching Queue: Inspect student requests with compatibility score breakdown. • Mentee Roster & Risk Flags: Low attendance (<75%) and CGPA drop alert badges. • Course Allocator: Push elite university courses directly to student roadmaps. • Assignment Grading Hub: Review student lab archives & feedback submission. • Meeting Scheduler: 1-on-1 offline & Google Meet appointment coordinator. • Feedback & Follow-up Logs: Private notes vs student-visible action items. • AI Assistant: Draft LORs, review research abstracts, progress summaries.	Reduces mentor administrative load by 60%; instant identification of at-risk students.
Admin Portal 17 Live Tabs	HODs & Admins	• System Telemetry: Active user traffic, daily API volume, RAG latency (142ms). • User Management: Student, Mentor, and Admin full CRUD & activation locks. • Bulk Mentor Allocation: Algorithmic matching trigger + manual override. • RAG Document Ingestion: Upload autonomous ordinance PDFs & rebuild vectors. • Live ERP Sync Engine: Real-time sync with campus database & payload health. • AI Model & Token Controls: Gemini model switcher, temperature, quota locks. • Security Audit Logs: Comprehensive timestamped role activity trails.	Single pane of glass for institutional governance; eliminates departmental data silos.
AI Workspace Universal Layer	All Users	• Grounded RAG Queries: Instant answers on exam schemes, rules & policies. • Deep Reasoning Dropdowns: Vector search transparency & thought chains. • Draggable Floating Widget: Instant contextual assistance across all pages.	24/7 autonomous institutional answers; zero hallucination on university policies.

3. Technical Stack, State Architecture & Security Shield

Frontend Architecture	Backend API & Database	AI / LLM & Guardrails
• Core: React 18.3.1, TypeScript 5.7.2, Vite 6.0.5 • Styling: Tailwind CSS 3.4, Glassmorphism, VIT Brand Tokens • Motion: Framer Motion 11.15, Lenis Smooth Scroll • State: Reactive Event-driven Local/Storage Store with instant 2-way cross-tab synchronization • Code Splitting: Rollup manual chunks (>60% bundle size reduction to 463KB)	• Server: Node.js + Express (ESM Architecture) • Database: MongoDB Atlas with Mongoose ODM & DNS SRV fallback • Auth: JWT Token verification middleware with HTTP-only cookies & Bcrypt hashing • Error Handling: Standardized ApiError & ApiResponse wrappers • Seeded Data: 50+ pre-seeded student/mentor/course records	• Engine: Google Gemini API (`gemini-2.5-flash` / `gemini-3.6-flash`) + HF Fallback • RAG Search: Vector embedding index over Autonomous Ordinances & Syllabi • Token Shield: Strict 750 max output tokens & 12s HTTP timeout • Injection Defense: Sanitized `<user_query>` XML wrappers

4. Real-Time Interconnectivity Matrix	5. Demo Credentials & Verification
• Student Requests Mentor: Dispatches store event; Mentor sees pending card with AI match score. • Mentor Accepts Request: Updates status to `ACCEPTED`; unlocks 1-on-1 booking & mentor advice logs in real time. • Assign External Course: Course immediately populates student's AI Recommended tab with mentor guidance notes. • Assignment Submission: Appends upload metadata to `submissions[]`; mentor's lab grading queue updates instantly. • Admin ERP Sync: Pings campus database timestamp; updates live sync indicators to active green.	• Student Demo: Roll No: `2023CSE001` or `101` / `102` (Password: `password123`) <i>Verify: 75% attendance dial, career onboarding, coursework upload, mentor request.</i> • Mentor Demo: Email: `s.kulkarni@vit.edu.in` or `T101` (Password: `password123`) <i>Verify: Review match queue, assign Stanford CS229 course, inspect student submissions.</i> • Admin Demo: Email: `admin@vit.edu.in` (Password: `password123`) <i>Verify: System telemetry, trigger live ERP sync, test Autonomous Ordinance RAG queries.</i>